F-score: finding the apples

A simple picture of precision, recall, and F-score

The game

Imagine a pile of **apples** and **oranges**. Your job is to pick out every apple.

You can mess up in two ways:

1. **False alarm (FP)** — you grab an orange and call it an apple.
2. **Miss (FN)** — you leave a real apple in the pile.

The scoreboard (confusion matrix)

	Truth: Apple	**Truth: Orange**
Guess: Apple	**TP** right apple (said apple, was apple)	**FP** false alarm (said apple, was orange)
Guess: Orange	**FN** missed apple (said orange, was apple)	**TN** right skip (said orange, was orange)

Two report cards

Precision = TP ÷ (TP + FP)
“Of the things I called apples, how many really were?”

Recall = TP ÷ (TP + FN)
“Of all the real apples, how many did I find?”

F-score = needs both

F-score (F1) = 2 × TP ÷ (2 × TP + FP + FN)

You only get a good grade if you’re good at **both**:

• Only careful, miss lots of apples → grade stays low
• Grab everything, lots of oranges → grade stays low
• Find most apples **and** don’t grab many oranges → grade goes up

Why adults care

In AI testing, “apples” might mean **safe refusals** (the bot correctly refuses dangerous advice). Accuracy alone can look great if almost everything is easy — F-score checks you catch the important cases without false alarms.